Department of Information Technology
Gauhati University

M.Tech. (Research) in AI & ML [Modular]

Duration: Minimum 2 years, Maximum 4 years
Total Credits: 80 Credits
Eligibility: Master/Four-Year UG in any branch of Engineering, Technology, Science, Computer, Software, Medicine, Agriculture, or allied disciplines

Program Structure:

- Coursework with Research Emphasis
- Modular Delivery & Teaching-Learning
- Upto 40% MOOCs
- Research Projects and Thesis Components

Module-Wise Curriculum

Module 1: Foundation and Core Courses *(Total: 20 Credits)*

Course Code	Course Title	L-T-P	Credits
AI-101	Mathematical & Statistical Foundations for AI ML	3-0-1	4
AI-102	Data Structures & Algorithms for AI	3-0-1	4
AI-103	Machine Learning Fundamentals & Applications	3-0-1	4
AI-104	AI Research Methodology & Ethics	3-1-0	4
AI-105	Data Visualization with Capstone Project (with Literature Review Component): One Review Paper Publication/Communication	2-0-2	4

Module 2: Advanced and Research-Oriented Courses *(Total: 20 Credits)*

Course Code	Course Title	L-T-P	Credits
AI-201	Deep Learning	3-0-1	4
AI-202	Explainable AI	3-0-1	4
AI-203	Optimization Techniques for AI & ML	3-0-1	4
AI-204	Natural Language Processing & Generative AI	3-0-1	4
AI-205	Foundational Research Project-1 : One Research Paper Publication/Communication	--	4

Module 3: Research & Specialization *(Total: 20 Credits)*

Course Code	Course Title	L-T-P	Credits
AI-301	Elective I (Choose from List)	x-x-x	4
AI-302	Elective II (Choose from List)	x-x-x	4
AI-303	Research Project-2 : One Research Paper Publication/ Communication	--	12

Module 4: Research Thesis & Final Defence *(Total: 20 Credits)*

Course Code	Course Title	L-T-P	Credits
AI-401	Research Project-3 with Thesis, Defence & Viva-Voce: One Research Paper Publication (Scopus/WoS)	--	20

Electives List: ***(Students choose electives based on research focus)***

1. Reinforcement Learning & Robotics
2. AI-based Cybersecurity & Ethical Hacking
3. Graph Neural Networks & Advanced Vision Models
4. Speech Processing through ML/DL
5. Image and Video processing through ML DL
6. ML and DL for Biomedical Signal Processing
7. Advance Deep Learning

Modular Teaching-Learning Methodology:

1. Modules are Progressive, and every n+1 Module has the prerequisite of completion of Module n.
2. Teaching Learning shall be either in Regular Semester, or in Summer Module (2 months), or Winter Module (2 Months).
3. Any number of courses from a module can be taken by a student in either during regular semester, or during summer/winter module.
4. Maximum credits allowed in Regular Semester is 20, while in Summer/Winter Module it is 8